D2W Hybrid Bonding Challenges for HBM

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Abstract

This paper addresses the key requirements for a successful Die-to-Wafer hybrid bonding process for die stacking. The paper delves into requirements, principles, challenges, and advancements made using integrated processing equipment. The process requires key surface preparation steps integrated with the Hybrid bonders to achieve the highest levels of yield. Unique challenges for memory stacking including thin die handling, alignment, front and backside surface preparation.

Keywords— Die to Wafer, Hybrid bonding, HBM

I. INTRODUCTION

This paper discusses the importance of Die to Wafer (D2W) hybrid bonding in the development of high bandwidth memory (HBM) for high-performance computing (HPC) and artificial intelligence (AI) applications. HBM capacity is crucial for fueling the computational intensity of these workloads. Currently, HBM packages stack DRAM dies using thermocompression or flip chip bonding with copper (Cu) pillars made from a combination alloy of Cu-Nickel-Tin Silver (Cu-Ni-SnAg). The space between the pillars is filled with epoxy mold compound to complete the stack. D2W hybrid bonding eliminates the need for Cu pillars by directly bonding the die surfaces of Cu and dielectric materials onto a target wafer. This process involves preparing the surfaces of both the substrate and target using plasma and wet cleaning processes, followed by using a high accuracy bonder to execute the chip pickup and bonding. The benefits of D2W hybrid bonding include improved performance, thermal dissipation, and reduced stack height.

After one layer of die is bonded to the target wafer the process can be repeated by re-preparing the surface of the target wafer followed by D2W bonding of another layer of DRAM chiplets thus building the stack.

II. MATERIALS TO SYSTEMS CO-OPTIMIZATION

We have implemented a simulation platform, referred to as Materials to Systems Co-Optimization (MSCO™), that is structured into four co-optimization loops [1]. The first loop considers materials, process, and interconnects. The die/channel loop considers an aggregate of interconnects to capture effects such as crosstalk, signal integrity, die warpage, etc. The design loop considers the electrical performance and thermo-mechanical reliability of the interposer, package, and PCB, including assembly process flow optimization.

Ab-initio material studies examined dielectric-dielectric bonding energies for several dielectrics [2], and the kinetic energy, incident angle and flux of the plasma surface treatment were found to correlate well with surface activation and improved dielectric-dielectric bonding [3]. The effect of copper grain size and distribution on copper CTE [2] and the effect of copper CTE on bonded pad thermo-mechanical stability and reliability versus lower hybrid bonding thermal budget [4] were also studied with simulation. A process simulation study showed improved post-CMP wafer flatness by eliminating barrier protrusion and reducing dishing/dielectric rounding through optimized fiducial mark design [5].

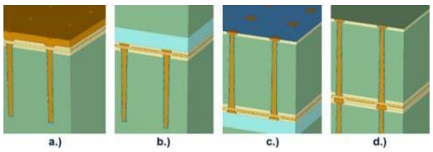

Figure 1. Through-process Structural Emulation Sequence: a.) TSV& BEOL Formation, Pad Copper Fill, b.) Carrier Wafer Bond, c.) Backside Copper Pad Formation, d.) Hybrid-bond Die Stack

Fig. 1 shows a through-process structural emulation that is calibrated to available TEMs. The calibrated hybrid-bonded HBM structure is used for thermal-mechanical simulations that are currently in progress.

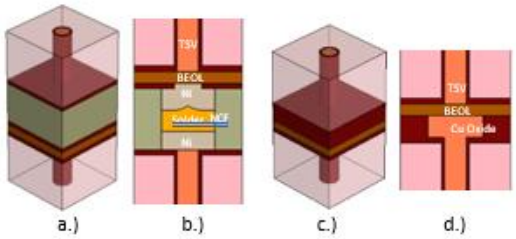

*Figure 2. a.) 3D structure with micro bump, b.) 2D cross-section of die with micro bump c.) 3D structure with hybrid bonding, d.) 2D cross-section of die with hybrid **bonding***

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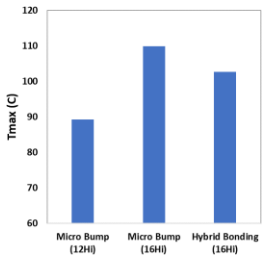

Figure 3. Maximum temperature in HBM dependence on stack number and bonding technology

We conducted thermal simulation to compare the thermal resistance for Silicon die with micro bump and hybrid bonding interconnect (Fig. 2). The thermal resistance for hybrid bonding interconnect is 47% lower compared to micro bump due to the interconnect thickness reduction. The third generation HBM 12Hi consists of one buffer die at the bottom and 12 stacked core dies, while the fourth generation is expected to be HBM 16Hi. As illustrated in Fig. 3, maximum temperature increases about 20°C when increasing the stack number from 12Hi to 16Hi. This is due to the increased power in the stack and the increased thermal resistance. Hybrid bonding with 47% lower thermal resistance compared to Micro bump, can reduce the maximum temperature by about 8°C for 16Hi. Power reduction and further optimization is needed to lower the temperature in HBM stack.

III. Hybird Bonding Process for HBM Stacking

A. *Process flow for HB stacking*

Fig.4 illustrates the process flow for HB stacking in Die-to-Wafer (D2W) Hybrid Bonding technology involves several steps and challenges. In multi-stack configurations, thinner device dies (<50um) are used, requiring both front side and back side connection pad processes. Through Silicon Via (TSV) formation is necessary to establish connections between the front and back sides. The test vehicle process flow for memory stacking is more complex than that for logic devices, considering TSVs, thin wafer handling, and the need for both front and back side damascene processes.

Multi-stack hybrid bonding requires temporary carrier (TBDB) technology to support thin wafers after the front side damascene process. However, the temporary carrier adhesive poses challenges, as it should not be exposed to elevated temperatures or vacuum processes during subsequent steps. To protect the front side surface, a thin sacrificial layer of coating material can be applied. The wafer is then thinned down, and the back side vias (TSVs) are exposed through a back side via reveal (BVR) process. Precise control of the BVR chemical mechanical planarization (CMP) process is crucial to prevent TSV copper penetration and breakage. A back side damascene process follows to create back side copper pads. After completing both front and back side processes, the carrier is detached from the wafer's front side, and singulation takes place. "Litho based plasma dicing" is a preferred method for die singulation, but it requires a clean dicing lane free from metal structures. Careful selection of tape adhesive is necessary to ensure easy cleaning and compatibility with the bonding surface. Overall, the process flow for HB stacking involves multiple steps and challenges, including thin wafer handling, TSV formation, back side damascene, and proper adhesive selection.

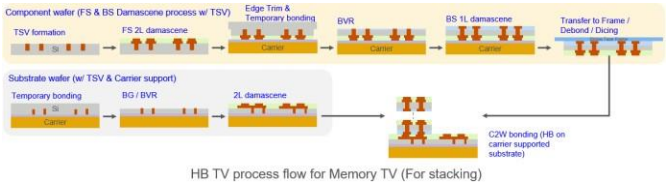

Figure 4. Test Vehicle process flow for Memory Stacking

B. *Challenges with Hybrid Bonding*

Die-to-Wafer (D2W) hybrid bonding offers numerous benefits, but it comes with several technological challenges that need to be addressed for successful integration. Common challenges include achieving optimal bonding by ensuring polished dielectric surfaces and controlling the Copper (Cu) chemical mechanical planarization (CMP) profile. Creating an optimal topography for bonding surfaces and protecting them during cleaning and activation steps is crucial (Fig 5). Proper pre-treatment of dielectric surfaces through plasma activation and hydration is necessary for hydrophilic bonding. Careful process conditions should be chosen to minimize material etching and sputtering. Cleanliness is vital for successful bonding, as particles can negatively impact performance by creating voids and increasing contact resistance. D2W hybrid bonding also presents unique challenges in pre-treating chiplets, which are obtained through singulation and attached to frames using plastic tapes. Exposure to the plasma environment can lead to the outgassing of volatile organic compounds (VOCs), contaminating surfaces. Achieving optimal electrical performance requires strong bonding and precise chiplet placement. Each die is individually picked and placed on substrate wafers, requiring precise alignment for thousands of placements. Successfully addressing these challenges is crucial for the successful integration of D2W hybrid bonding and realizing its potential benefits.

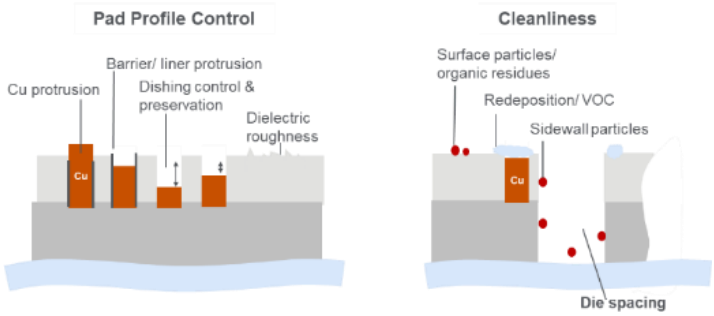

Figure 5. Pad profile control and cleanliness illustrations for HB

Process Co-optimization

Co-optimization is performed to achieve ideal planarization and topography in hybrid bonding. This involves considering processes such as PVD barrier, Cu liner, ECD, and Cu CMP. The thermal expansion properties of different materials must be considered as they can affect surface topography during thermal processing. It is crucial to set the heights of each material properly to accommodate height changes during the bonding anneal step. The goal is to achieve optimal bonding without any barrier or Cu protrusion after CMP. The dishing amount, which refers to the removal of material from the surface during CMP, should be limited to approximately 5nm or less. This ensures that the bond pads remain in contact with each other during the thermal expansion caused by the post-bonding anneal (PBA). By achieving optimal planarization and controlling topography, successful hybrid bonding can be achieved.

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Surface Activation

In the pursuit of optimal surface topography, further investigations were conducted on dielectric surface activation, the impact of a particle-free surface, and the effect of queue time in Die-to-Wafer (D2W) hybrid bonding. These requirements are more demanding in D2W bonding compared to Wafer-to-Wafer (W2W) bonding due to the use of flexible organic tape or carrier wafers with organic adhesive during the processing of dicing and singulated dies. Plasma activation has proven effective in enhancing the hydrophilicity of dielectrics like SiO2 or SiCN by introducing silanol groups (Si-O-H) to the surface[9]. These silanol groups facilitate initial bonding between substrates and chiplets, even at room temperature. The attachment of these groups can be achieved through ion energy, which introduces disorder and high-energy states to the surfaces. The hydrophilicity of dielectric surfaces can be evaluated by measuring water contact angle (CA), with superhydrophilicity (CA < 5°) achievable under appropriate plasma conditions for SiO2 surfaces. However, exposure to air after activation gradually increases the contact angle, leading to poor bonding performance and delamination at the die edges (figure 6a.). Thus, precise control of the queue time between activation and bonding is crucial for optimal bonding. Selecting the right plasma conditions, including ion energy, density, and chemistry, is critical to achieving sufficient activation without causing physical or chemical damage to materials like Cu and plastic tapes. High ion energy levels can result in roughened dielectric surfaces, leading to voids and compromised bonding performance (figure 6b.).

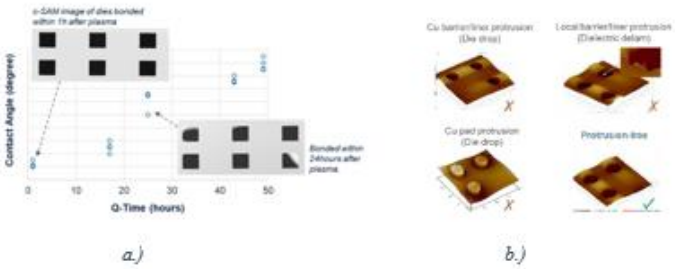

Figure 6. a.) CSAM data post bond showing sensitivity of bond quality to queue time b.) AFM data showing optimized pretreatment process without protrusions in the liner or Cu pads

Defectivity Control

Maintaining cleanliness is crucial for optimal performance in hybrid bonding. Even the presence of small particles can lead to bonding issues and failure[10]. Therefore, strict control over particle occurrence is necessary, and activation chambers should meet device specifications for effective particle control. A thorough wet cleaning process is indispensable to ensure a particle-free surface before bonding. However, achieving the required cleanliness level for diced wafers on flexible tapes can be challenging. The dicing process itself can introduce additional particles and contaminants. Insufficient cleaning after dicing can leave particles on the tape or die sidewalls, which can dislodge and land on the die surfaces, creating voids near the edges (Fig 7.).

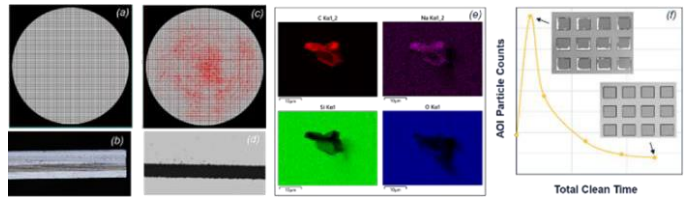

Figure 7. Automatic optical inspection (AOI) of the surface (a) and sidewall (b) of dies after dicing. Many particles (black) can be seen adhering to the sidewall. Unoptimized cleaning redeposits the particles onto the die surface (c-d). Particles detected by energy- dispersive X-ray spectroscopy indicate tape adhesive as their source. If left unremoved, these particles can lead to die delamination, as observed in SAM inspection. (f)Bonder solutions for Stacking

Hybrid bonding at wafer level has been around for many years but D2W HB faces issues not just in the preparation phase but also at the bonding stage. One critical element in HB is the bond front propagation. As the ultra-clean faces of die and substrate wafer come into contact, Van der Waal's forces trigger, and the 2 faces are drawn to each other. Two main factors need to be addressed: the contact area at which this happens and the speed at which this occurs.

There should only be one bond front propagation during bonding. Several bond fronts will likely result in air entrapment leading to voids if they collide [6, 7]. A single point of contact is needed when the die touches the substrate wafer. This could be a corner, an entire edge, or the center of the die, as seen in Figure 8. The latter is the method that is most applicable to D2W bonding since the former two require manipulation of the die in x, y, and z whilst compensating for theta, which is something die bonders are not designed for, nor efficient enough to target throughput levels due to excessive manipulation.

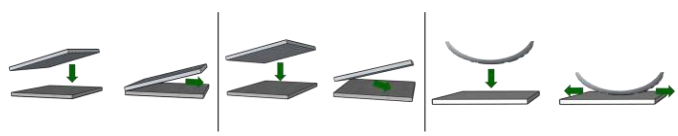

Figure 8. Bonding schemes for bond front propagation control: (a) initial contact at one edge, (b) initial contact at one corner, (c) initial contact in the middle of the die after shaping.

This method of die shaping however faces some challenges such as die thickness, die size, and die aspect ratio amongst others. Thicker dies are obviously more resistant to die shaping whereas small dies pose similar issues as well as difficulty in handling and shaping. Die aspect ratios are a challenge not existing in W2W bonding, since whatever wafer size, the shape is always round. The issue with dies having high aspect ratio is that at some point (Figure 9d) the bond front starts occurring only unilaterally along the long side of the die which can give rise to a die scaling effect.

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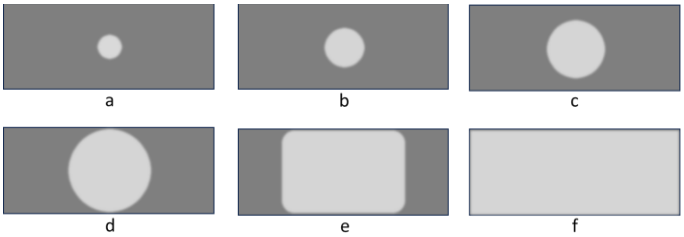

Figure 9. Top view of assumed bond front propagation on high aspect ratio die on substrate wafer (as would be visible through IR) from (a) at time of contact to (f) after full bond. The dark grey area denotes the unbonded section, and the light grey area is the bonded section.

Bond front propagation therefore impacts void formation through air entrapment, void formation, or edge delamination through moisture formation [7], as well as die scaling (stretching or contracting). These effects play a more vital role when stacking dies, and the higher the stack the higher the impact magnitude. This is further exacerbated in the memory scenario, where stacks upwards of 16Hi are being targeted.

By the nature of HB itself, various die material combinations can be bonded together (Si, GaAs, GaN…) as well as different dielectrics (SiO2, SiCN, SiC…) and respective thicknesses. As such die handling tooling must be designed to cater for these criteria. In an ideal world one set of tooling can address all the above possible combinations, but this is not possible. As such a tool which can provide a degree of freedom in die shaping has been designed whereby for the same die size, different degrees of shaping can be obtained by varying a parameter. The results shown in Figure 10. (right) compare the die shaping height for 2 silicon dies at 50µm and 100µ thickness. The degree of shaping height might vary for different die materials and dielectrics. The illustration in Figure 10. (left) shows alternative tool designs which can accomplish die shaping but do not offer a dynamic adjustment. For certain applications these tools may provide better results. Various designs and dimensions of the pedestal providing the die shaping have been designed, manufactured, and tested.

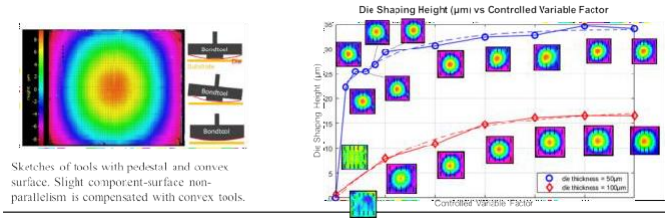

Figure 10. Sketches of tools with pedestal and convex surface (left), and maximum warpage height for 50µm and 100µm thick dies at different die shaping parameters (right).

The process flow for die stacking requires that the partially populated wafer exits and enters the bonder for wet cleaning, plasma activation and hydration. This poses two factors to be addressed: first, the populated substrate wafer now has a different warpage arising from the bonded dies on it, leading to potentially different heights across the wafer, and second the bonding height has now physically changed, even if in the µm regime, from both the warpage as well as due to the fact that the wafer was physically unclamped from the chuck and removed before being loaded again. The bonder has, therefore, to be able to accurately determine the new point of contact in z axes to be able to perform a successful bond. A solution to address this was thus developed to cater for this is by means of advanced algorithms coupled to the development of advanced optical hardware and sensors around and in the bond head.

IV. Acknowledgements

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